

# CSC633 MACHINE LEARNING



TEXAS A&M UNIVERSITY

Department of Computer  
Science & Engineering

# CSCE 633: Machine Learning

## Lecture 1: Introduction to Machine Learning

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# Topics

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- Introduction to Machine Learning
  - What is machine learning?
  - Models and accuracy
  - Some general supervised learning examples
  - Terminology!

# Standard Machine Learning Terms (1)

## Tasks

- Machine Learning tasks typically fall into the following tasks:
  - **Classification** - assigning a category to each item  
i.e. text classification, speech recognition, cats vs. dogs
  - **Regression** - learning a real value for each item  
i.e. stock value prediction
  - **Clustering** – learning how to partition a set of items into homogenous subsets i.e. identifying communities within larger groups of people
  - **Dimension Reduction** - learning a transformation from the initial representation into a lower-dimensional representation i.e. digital image processing

## Terminology

# Standard Machine Learning Terms (2)

## Tasks

- Machine Learning tasks typically fall into the following tasks:
  - **Classification** - assigning a category to each item i.e. text classification, speech recognition, cats vs. dogs
  - **Regression** - learning a real value for each item i.e. stock value prediction
  - **Clustering** – learning how to partition a set of items into homogenous subsets i.e. identifying communities within larger groups of people
  - **Dimension Reduction** - learning a transformation from the initial representation into a lower-dimensional representation i.e. digital image processing

## Terminology

- Key terms and definitions common in machine learning
  - **Examples** - instances of data for learning or evaluation
  - **Features** – set of attributes representing the examples (often a vector)
  - **Labels** - values/categories assigned to the examples
  - **Hyperparameters** – free parameters of a model not determined in training but rather specific as inputs
  - **Training sample** – examples used in the training process
  - **Validation sample** – examples used in parameter tuning
  - **Test sample** - examples used for performance evaluation
  - **Loss function** – a function measuring the difference between a predicted label and the true label. This function is to be minimized in the optimization stage
  - **Hypothesis set** – a set of possible functions which map the feature vectors to the set of labels

# Other Learning Scenarios

- **Online Learning** – training and testing phases are intermixed. The data is made available to the learner over time. The objective is to minimize the cumulative loss over all rounds.
- **Reinforcement Learning** – training and testing phases are intermixed. The learner interacts with/affects the environment. Some actions of the learner lead to being rewarded. Objective is to maximize the rewards based on a course of actions. Exploration vs. exploitation dilemma.
- **Self-Supervised Learning** – the learner interactively collects training examples, querying an oracle for requesting new labels. Used when labels are expensive/difficult to obtain.

# Takeaways and Next Time

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- What are machine learning tasks?
- What are some basics of machine learning?
- Can a learner always achieve zero prediction error?
- Next session: Review of Linear Algebra, Measures of Model Performance, Linear Regression and K-Means